Aryamaan Dash

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EDUCATION

University of California, Irvine — Sep 2025 - June 2028
B.S. in Computer Science and Engineering, Mathematics Minor, GPA: 3.6 — Irvine, CA

- **Coursework**: Python Programming with Libraries, C++ as Secondary Language, Intermediate Python Programming, Digital Logic Laboratory

EXPERIENCE

Hey, Blue! — Mar. 2024 – Dec. 2024
Front-End Developer — ONLINE

- Website developer with ReactJS, HTML, CSS
- Worked with a team of website developers as a developer and as a team lead.
- Created multiple pages, added multiple features, and distributed workload proportionally.
- 15 hours per week during non-academic weeks, 8 hours per week during the academic year.

Mathnasium — Sep 2022 – May. 2024
In-Center Instructor — Livermore, CA

- Mathematics instructor at a tuition center teaching students from ages 5 to 16.
- Taught math students topics from counting to calculus and helped students with homework.
- Taught approximately 5 to 10 different students per week.
- Managed a database of grades and assigned work based on weak points.

PROJECTS

Study Tracker

- Developed a full-stack study tracking web application using Next.js, TypeScript, Prisma Postgres, and Vercel to help users log, manage, and review study sessions.
- Built interactive data visualizations with Chart.js to display study trends, session history, and progress insights in a clean, user-friendly dashboard.
- Designed and integrated a persistent database-backed tracking system with Prisma Postgres, enabling reliable storage and retrieval of user study data.

RISC-V RV32I Single-Cycle Processor

- Designed and implemented a single-cycle processor in Verilog using Vivado, integrating core datapath components including the program counter, register file, ALU, immediate generator, instruction memory, and data memory.
- Developed Verilog testbenches to verify individual modules and processor-level functionality, validating instruction execution, control signals, register updates, ALU operations, and memory behavior.

Programmable Multi-Effects Guitar Pedal

- Built a Daisy Seed-based programmable guitar pedal with selectable bypass, distortion, reverb, delay, and flanger effects.
- Programmed real-time embedded C++ DSP firmware using DaisySP, audio callbacks, delay-line processing, wet/dry mixing, and debounced mode switching.
- Soldered a perfboard prototype integrating guitar input buffering, 3.3 V signal biasing, audio jack wiring, control inputs, and analog/digital ground connections.

Guitar Audio Classification Model

- Built an audio classification pipeline with 89% test accuracy to identify four guitar sound categoriessingle notes, chords, palm-muted playing, and background noisefrom self-recorded audio.
- Processed recordings into one-second WAV clips and extracted MFCCs, RMS energy, spectral centroid, bandwidth, rolloff, and zero-crossing-rate features for model training.
- Trained and evaluated a logistic regression classifier using session-level training, validation, and test splits to prevent data leakage and measure performance on independently recorded guitar samples.

TeachBack: Inverted Tutoring

- Started at an AI for Education Hackathon and built an inverted tutoring app where the AI pretends to be the student while the real student teaches it.
- Designed the AI flow to use EverOS memory to track what each student struggles with and personalize future explanations.
- Built step-gated problem solving, answer validation, and a Butterbase-ready backend schema for learners, practice sets, progress, and tutor messages.

SKILLS

Languages : Python, TypeScript, HTML/CSS, C++, Verilog
Tools : Git, Vivado, Next.js, ESP32, Machine Learning Models, Daisy Seed, Soldering